AMPRON

RoHS

SPECIFICATION FOR APPROVAL

承 认 书

客户名称（CUSTOMER）：	Dfind
产品名称（PRODUCT ITEM）：	NTC 热敏电阻器（NTC THERMISTOR ）
客户料号 (CUSTOMER PART NO.)：	
规格型号 (PART NO.)：	MF52E-503H3950FAL36 脚漆
文件编号 (FILE NO.)：	MO190160
版　本（EDITION）：	A/0
编制日期（DATE）：	2019-08-14

客户确认(CUSTOMER CONFIRM)
样品确认后请在一周内回签，谢谢！

供货商确认 (SUPPLIER CONFIRM)	
制 作（Edited）	梁　芳
审 核（CHECK ）	肖海艳
批 准 （APPROVE）	刘　平
公司签章	

公司名称	
■ **深圳安培龙科技股份有限公司**	**Shenzhen ampron technolody co.,Ltd.**
□ 深圳市安培龙敏感技术有限公司	Shenzhen AMPRON Sensitive Components Co., Ltd.
公司地址：深圳市龙岗平湖街道平湖社区富民工业区富康路 43 号 65 号厂房一至四楼 Add：Floor 1-4,Building 65, FuKang Road ,43#, FuMin Industrial Park, Pinghu Town,Longgang District,518111, Shenzhen , Guangdong, China	
TEL：（86）-0755-84012223	FAX：（86）-0755-84012280
Http:/www.ampron.com	E-mail:sales@ampron.com
制造基地	
东莞市安培龙电子科技有限公司	**Dongguan AMPRON electronic technology Co., Ltd.**
工厂地址：中国.广东.东莞市清溪镇厦坭村金星工业区南区	
Address：Southern area ,Jinxing industrial estate ,xiani village ,qingxi town,dongguan city , guangdong province, China	
TEL：（86）-0769-87895315	FAX：（86）-0769-87895319
郴州安培龙传感科技有限公司	**CHENZHOU AMPRON Sensing TechnologyCo.,Ltd**
Address：湖南省郴州市桂阳县工业园长富项目区	
Chang Fu project area of Guiyang Industrial Park, Chenzhou City, Hunan Province	
TEL：（86）-0735-4816881	FAX：（86）0735-4816883

0. 修订履历表

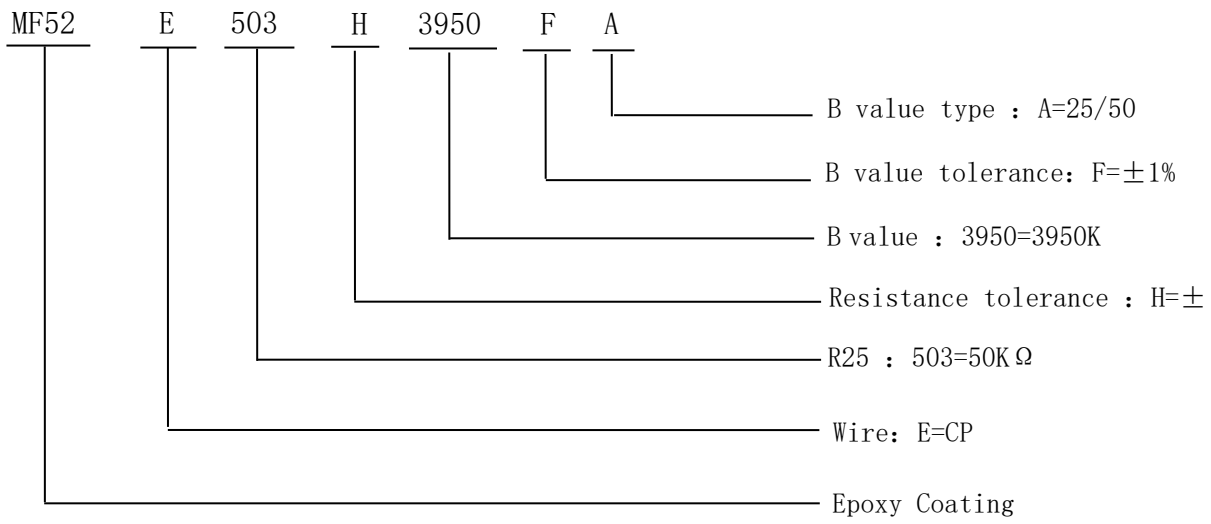
修订条款	修订前内容	修订后内容	修订日期	修订前版本	修订后版本	修订人
全部	首次发行		2019-08-14		A	梁 芳
	以下空白					

1.Use

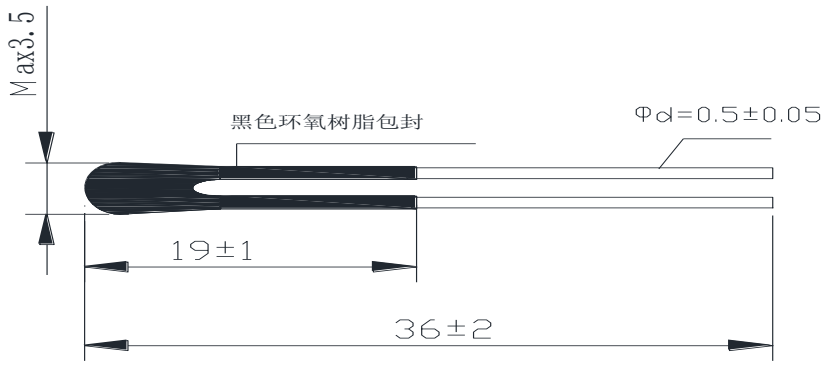
The Specification is suitable for NTC Thermistor.

This product comply with European Union ROHS directive requirements.

2. Making of Part Number：



3. Shape, Size（unit: mm）



No	Part Name	ROHS
①	Black Expoxy	PASS
②	CP Wire	PASS

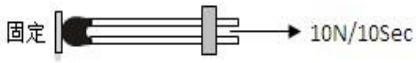
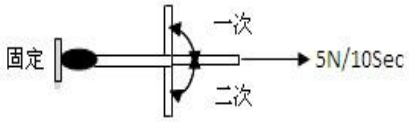
4. Electrical Characteristic

NO.	Item	Condition	Technical requirements
4-1	Resistance	25℃ ,At=25℃ ±0.05 ℃ 50℃ ,At=50℃ ±0.05 ℃	R_{25}＝50KΩ±3% R_{50}＝17.941KΩ（参考值）
4-2	B Value	B value between 25℃ and 50℃ B=Ln(R1/R2)/(1/T1-1/T2) T1 = (273.15 + 25) K T2 = (273.15 + 50) K Remark:273.15 is absolute temperature	$B_{25/50}$＝3950K±1%
4-3	Working Temperature Range	The temperature range under the stated use conditions	T_W＝－40℃～＋125℃
4-4	Thermal Time	In the zero power conditions ,Still in the air， Thermistor temperature dropped to its initial temperature and eventually of 63.2% of the difference between the time required, general definition of 25 ℃ ~ 50 ℃	≤12 seconds
4-5	Hit Dissipation Factor	In the 25 ℃ in the still air, make the thermistor temperature increases 1 ℃ for the power consumption	δ ≥3mW/℃
4-6	Rated Power	25℃ power rating	≤50mW
4-7	Insulation resistance	At The Usual Temperature Pressure 500 VDC and maintain 60 SEC	≥100MΩ

5. Reliable Test

NO.	Item	Standard	Method of Test	Technical Requirements
5.1	Solder Ability	IEC60068-2-20 (GB2423.28)	The lead wire shall be dipped into solder bath of 235±5℃ for 2±0.5S with 2-2.5mm space from the body Check the appearance after testing	more than 95% solderability

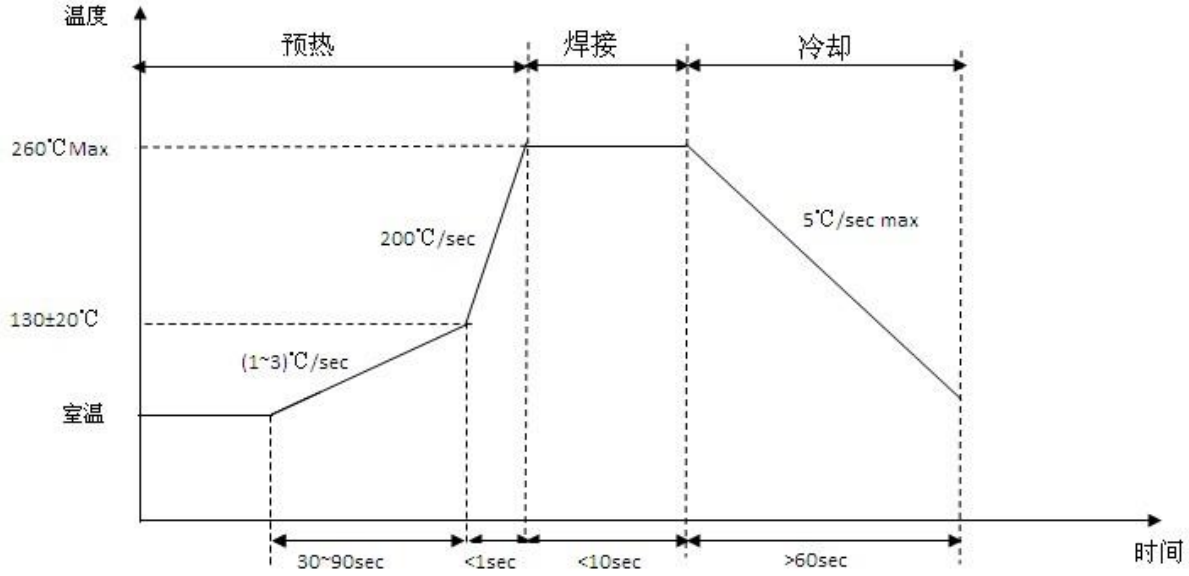
5.2	Soldering heat	IEC60068-2-20 (GB2423.28)	The lead wire shall be dipped into solder bath of 260±5℃ for 10±1S with 2-2.5mm space from the body, then be returned to and stabilized at room ambient. When the test finished, in room temperature often wet back 1 ~ 2 Hr, check the appearance and reiteration R25 rated zero power resistance	No obvious damage $\Delta R/R_{25} \leq \pm 3\%$

5.3	Lead full strength	IEC60068-2-21 (GB2423.29)	Test Ua: Pull 10N,lasting 10S;  Test Ub: Bending 90 degree, pull 5N,twice;  When the test finished, in room temperature often wet back 1 ~ 2 Hr, check the appearance and reiteration R25 rated zero power resistance	No obvious damage $\Delta R/R25 \leq \pm 3\%$
5.4	High temperature test	IEC60068-2-2 (GB2423.2)	Ambient Temperature:100±5℃， Time of Duration: 1000±24h When the test finished, in room temperature often wet back 1 ~ 2 Hr, check the appearance and reiteration R25 rated zero power resistance	$\triangle R/R25 \leq \pm 3\%$
5.5	Low temperature test	IEC60068-2-1 (GB2423.1)	Ambient Temperature :-40±5℃，Time of Duration : 1000±24h， When the test finished, in room temperature often wet back 1 ~ 2 Hr, check the appearance and reiteration R25 rated zero power resistance	$\triangle R/R25 \leq \pm 3\%$
5.6	Normal temperature electricity test		25±5℃，Power on 1000±24h,DC0.2mA。	$\triangle R/R25 \leq \pm 3\%$

5.7	Moisture Proof Test (Steady-state damp and hot)	IEC60068-2-3 (GB2423.3)	Environment temperature：40℃±2℃ Environment humidity: 90%～95% Duration：1000±24h When the test finished, in room temperature often wet back 1 ~ 2 Hr, check the appearance and reiteration R25 rated zero power resistance	△R/R25≤±3%
5.8	Temperature cycle	IEC60068-2-14 (GB2423.22)	Low Temperature：-30℃±2℃ Low Temperature Retention Time：3min±0.5min High Temperature：100℃±5℃ High Temperature Retention Time：3min±0.5min Low Temperature Retention Time：≤30S，Repeat 100 times When the test finished, in room temperature often wet back 1 ~ 2 Hr, check the appearance and reiteration R25 rated zero power resistance	△R/R25≤±3%

Remark1：With more performance test of the relevant supplementary conditons :Normal Temperature often wet ＝23℃±10℃、Relative Humidity:25%～85%。

6. Wave soldering schemes



7. Manual welding

Item	Condition
Welding temperature	330℃（max）
Welding Time	3 sec（max）
Contact Distance	6mm（min）

8. Test result record.

8.1 Item No

8.2 Lot No

8.3 Quantity，Delivery date。

8.4 Sampling observation data of R25、B value should be offered when delivery.。

8.5 Size Data should be offered when delivery.

9.Packing

9.1 Packing of products

500pcs（Weight exchange）is unit，Packed by plastic bag，Vacuum.（The below label just for reference）

NTC THERMISTORS	
Item：	Item No：
R25 ：	B value 25/50：
Qty：	Test Result：
Date：	Remark：

9.2 Delivery·Packing

Choose the packed cartons according to delivery quantity in order to avoid any damage during the transportation.。

9.3 Storage Condition

9.3.1 Storage Temperature：-10℃ ～ +40℃

9.3.2 Relative Humidity：≤75% RH

9.3.3 Avoid stored in corrosive and lighting environment.

10. Product certification

E243011 CQC13001089457 TÜVRheinland R 50301960 0001

11. Attention

Chip thermistor may be damaged , please obey the following matters:

11.1Thermistor is designed based on the specified purposes, don't use for others.

11.2If used for sensor, it needs to be assessed the reliability test and confirmed that no abnormal before using.

11.3Don't give thermistor exceed its maximum load power circuit voltage.

11.4. Don't use exceed its maximum temperature range.

11.5 Please do not used in the following environment:

A. Corrosive atmosphere（C12、NH3、SOX、NOX etc）

B. High conductivity of medium (electrolyte, water, brine, etc.)

C. Acid, alkali, organic solvents

D. The palace with much dust.

12. R-T 表

R25=50KΩ±3%			B25/50=3950K		料号：M056-5		版本：A
T(℃)	Rmin(KΩ)	Rnom(KΩ)	Rmax(KΩ)	T(℃)	Rmin(KΩ)	Rnom(KΩ)	Rmax(KΩ)
-40	1569.605	1675.991	1787.977	-3	182.840	191.039	199.425
-39	1467.665	1566.079	1669.588	-2	173.491	181.174	189.027
-38	1373.680	1464.812	1560.584	-1	164.685	171.887	179.244
-37	1286.868	1371.336	1460.033	0	156.372	163.126	170.018
-36	1206.540	1284.899	1367.116	1	148.525	154.859	161.318
-35	1132.095	1204.844	1281.113	2	141.116	147.058	153.113
-34	1062.994	1130.582	1201.387	3	134.120	139.696	145.373
-33	998.763	1061.599	1127.373	4	127.513	132.747	138.071
-32	938.982	997.435	1058.574	5	121.274	126.187	131.182
-31	883.273	937.679	994.540	6	115.380	119.994	124.681
-30	831.302	881.966	934.876	7	109.813	114.147	118.546
-29	782.766	829.968	879.224	8	104.552	108.626	112.756
-28	737.395	781.389	827.263	9	99.582	103.411	107.290
-27	694.944	735.965	778.705	10	94.886	98.486	102.130
-26	655.194	693.455	733.291	11	90.447	93.833	97.258
-25	617.943	653.643	690.783	12	86.251	89.437	92.657
-24	583.012	616.332	650.969	13	82.284	85.283	88.312
-23	550.235	581.341	613.653	14	78.534	81.358	84.207
-22	519.461	548.509	578.659	15	74.987	77.647	80.330
-21	490.553	517.685	545.825	16	71.633	74.140	76.666
-20	463.384	488.732	515.002	17	68.461	70.824	73.203
-19	437.839	461.525	486.054	18	65.459	67.689	69.931
-18	413.811	435.948	458.856	19	62.619	64.723	66.837
-17	391.201	411.895	433.293	20	59.932	61.918	63.912
-16	369.919	389.268	409.259	21	57.389	59.265	61.146
-15	349.881	367.974	386.654	22	54.981	56.754	58.531
-14	331.009	347.931	365.389	23	52.702	54.378	56.056
-13	313.230	329.059	345.378	24	50.544	52.129	53.715
-12	296.477	311.287	326.543	25	48.500	50.000	51.500
-11	280.688	294.547	308.812	26	46.406	47.862	49.319
-10	265.805	278.775	292.115	27	44.412	45.826	47.242
-9	251.773	263.914	276.391	28	42.514	43.886	45.262
-8	238.541	249.907	261.580	29	40.707	42.039	43.375
-7	226.061	236.705	247.626	30	38.986	40.279	41.577
-6	214.290	224.258	234.478	31	37.346	38.601	39.862
-5	203.185	212.522	222.089	32	35.784	37.002	38.227
-4	192.708	201.456	210.411	33	34.295	35.477	36.667

R25=50KΩ±3%			B25/50=3950K		料号：M056-5		版本：A
T(℃)	Rmin(KΩ)	Rnom(KΩ)	Rmax(KΩ)	T(℃)	Rmin(KΩ)	Rnom(KΩ)	Rmax(KΩ)
34	32.875	34.022	35.178	71	7.983	8.378	8.785
35	31.521	32.635	33.758	72	7.711	8.096	8.492
36	30.230	31.311	32.402	73	7.450	7.825	8.210
37	28.998	30.048	31.107	74	7.200	7.564	7.939
38	27.823	28.842	29.871	75	6.959	7.313	7.679
39	26.701	27.690	28.690	76	6.727	7.072	7.428
40	25.631	26.591	27.562	77	6.504	6.840	7.187
41	24.608	25.540	26.484	78	6.289	6.616	6.954
42	23.632	24.537	25.453	79	6.083	6.401	6.730
43	22.699	23.577	24.468	80	5.884	6.194	6.515
44	21.808	22.661	23.526	81	5.693	5.995	6.307
45	20.956	21.784	22.625	82	5.509	5.803	6.107
46	20.142	20.946	21.763	83	5.331	5.618	5.914
47	19.363	20.144	20.938	84	5.161	5.440	5.728
48	18.619	19.377	20.149	85	4.996	5.268	5.549
49	17.906	18.643	19.393	86	4.838	5.102	5.377
50	17.225	17.941	18.670	87	4.685	4.943	5.210
51	16.573	17.269	17.977	88	4.538	4.789	5.050
52	15.949	16.625	17.313	89	4.396	4.641	4.895
53	15.352	16.008	16.677	90	4.259	4.498	4.746
54	14.780	15.417	16.068	91	4.127	4.360	4.601
55	14.232	14.851	15.484	92	4.000	4.227	4.462
56	13.707	14.309	14.924	93	3.877	4.098	4.328
57	13.204	13.789	14.387	94	3.759	3.974	4.199
58	12.722	13.291	13.872	95	3.645	3.855	4.074
59	12.260	12.813	13.378	96	3.534	3.739	3.953
60	11.818	12.355	12.904	97	3.428	3.628	3.836
61	11.393	11.915	12.450	98	3.325	3.520	3.723
62	10.986	11.493	12.013	99	3.226	3.416	3.614
63	10.595	11.088	11.594	100	3.130	3.316	3.509
64	10.220	10.700	11.192	101	3.038	3.219	3.407
65	9.860	10.327	10.805	102	2.948	3.125	3.309
66	9.515	9.968	10.434	103	2.862	3.034	3.214
67	9.183	9.624	10.078	104	2.779	2.947	3.122
68	8.865	9.294	9.735	105	2.698	2.862	3.033
69	8.559	8.976	9.406	106	2.620	2.780	2.947
70	8.265	8.671	9.089	107	2.545	2.701	2.864

R25=50KΩ±3%			B25/50=3950K		料号：M056-5		版本：A
T(℃)	Rmin(KΩ)	Rnom(KΩ)	Rmax(KΩ)	T(℃)	Rmin(KΩ)	Rnom(KΩ)	Rmax(KΩ)
108	2.472	2.624	2.784	145	0.919	0.985	1.056
109	2.401	2.550	2.706	146	0.897	0.962	1.030
110	2.333	2.479	2.631	147	0.875	0.939	1.006
111	2.267	2.409	2.558	148	0.854	0.916	0.982
112	2.204	2.342	2.488	149	0.833	0.894	0.959
113	2.142	2.277	2.419	150	0.813	0.873	0.936
114	2.082	2.215	2.353				
115	2.024	2.154	2.289				
116	1.969	2.095	2.227				
117	1.914	2.038	2.167				
118	1.862	1.983	2.109				
119	1.811	1.929	2.053				
120	1.762	1.877	1.998				
121	1.714	1.827	1.945				
122	1.668	1.778	1.894				
123	1.624	1.731	1.844				
124	1.580	1.685	1.796				
125	1.538	1.641	1.749				
126	1.498	1.598	1.704				
127	1.458	1.556	1.660				
128	1.420	1.516	1.617				
129	1.383	1.477	1.576				
130	1.347	1.439	1.535				
131	1.312	1.402	1.496				
132	1.278	1.366	1.459				
133	1.245	1.331	1.422				
134	1.214	1.298	1.386				
135	1.183	1.265	1.352				
136	1.153	1.233	1.318				
137	1.124	1.202	1.285				
138	1.095	1.172	1.254				
139	1.068	1.143	1.223				
140	1.041	1.115	1.193				
141	1.015	1.088	1.164				
142	0.990	1.061	1.136				
143	0.966	1.035	1.108				
144	0.942	1.010	1.082				